



Department of Electrical and Electronics Engineering Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. EEE

Course Code & Title: EE3302 Digital Logic Circuits

Name of the Faculty member: Mr. A. Guna, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit I / Number System
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 1
- **Activity Chosen:** Think-Pair-Share
- **Justification:**

Number System Conversion

Four types of Number system conversion were taught. Since this concept students already studied in the higher secondary class, many studied felt easy in understanding the concept. Explanation about the conversion is given for 25 Minutes and this activity will greatly help me to know about the understanding levels of the students.

Time Allotted for the Activity: 25 Minutes

- **Details of the Implementation:**

After teaching the concept, the students were asked to form pair

Total Strength is 64, Number of Pairs – 32

Photographer: one student – Mr. R. Geothisman (interested in photography)

Report Prepared by me.

At the end the Class (Last 25 minutes)

- Students were asked to think about all the type of number system conversion for 5 minutes.
- Then Students were asked to share their share about number system conversion with their teammate for another 5 minutes.
- Then the selected pairs were asked to share their points in front of the classroom (5 Minutes)
- Students from other pair were asked to share the points which was missed by the presenting pair (5 Minutes)
- Finally, the number system conversion procedure of all the types were summarized (5 Minutes)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO9	PO10	PSO2
C202.1	3	2	1	1	1

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO10	PSO2
	3	2	1	1	1
Justification for correlation	Students apply fundamental mathematical concepts to solve number system conversion problems.	Students apply necessary conversion procedure for solving the given problem.	Students discuss with their team mate and work as a team	Students discuss with their team mate and effectively communicate their ideas	Students will be able to design and test the electronic system in the engineering applications.

- **Images / Screenshot of the practice:**

A view of implementation of the Activity while the students are sharing the points





- **Reflective Critique:**

- ❖ ***Feedback of practice from Students.***

- ✓ Students felt that they understood the concepts very easily when they discuss and learn with their friends.
- ✓ Students felt that activity was very useful and interesting and help to remember the concepts easily

- ❖ ***Benefit of the practice:***

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- Learning was made very interesting and the students actively participated in this event.
- The success of the activity was evaluated by asking the same question Assignment 1– Around all the students answered correctly.

❖ **Challenges:**

- Few Students hesitated to present their points in front of the classroom
- Late Bloomers were not able to understand some topics during discussion hours.
- Duration of this activity is not sufficient, since many students want to share their ideas

References:

- ❖ <http://www.readwritethink.org/professional-development/strategy-guides/using-think-pair-share-30626.html>
- ❖ <https://in.video.search.yahoo.com> – Think pair share

Signature of Faculty Member

HOD



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Innovative Practice Description

- **Unit / Topic:** Unit III / Synchronous Sequential Circuits
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen:** One-minute paper
- **Justification:**

Types of Flip-Flops

Four types of Flip-Flops were taught. Truth Table, Characteristic Table and Excitation table of all the four types of flip-flops were explained to the students for 45 Minutes and this activity will greatly help me to know about the understanding levels of the students.

Time Allotted for the Activity: 10 Minutes

- **Details of the Implementation:**

After teaching the concept, the students were asked to write the learnt concepts in a sheet

Photographer: one student – Mr. R. Geothisman (interested in photography)

Report Prepared by me.

At the end the Class (Last 10 minutes)

- Students were asked to think about all the type of Flip-Flops (5 minutes)
- Then Students were asked to write their points in a sheet (1 minute)
- Finally, all the types of flip-flops were summarized (4 Minutes)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO10	PSO2
C202.3	3	2	1	1

- PO / PSO mapped:

Innovative practice	PO1 3	PO2 2	PO10 1	PSO2 1
Justification for correlation	Students apply fundamental mathematical concepts to understand the type of input given to Flip-Flops	Students identify the necessary data of the flip-flop for solving the flip-flop conversion problem.	Students comprehend and write effective report about the learnt concept	Students will be able to design and test the electronic system in the engineering applications.

- Images / Screenshot of the practice:

A view of implementation of the Activity while the students are sharing the points

EE3302
JK Flip flop. 953621105018
G. Angalaeswari

Truth table.

clk	J	K	Q _{n+1}
0	X	X	Q _n
1	0	0	Q _n
1	0	1	0
1	1	0	1
1	1	1	Q _n

characteristic table.

Q _{n+1}	J	K	Q _n
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Excitation Table.

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

SR FLIP FLOP M. Jeyarajalakshmi
953621105026

TRUTH TABLE

clk	S	R	Q _{n+1}
0	X	X	Q _n
1	0	0	Q _n
1	0	1	0
1	1	0	1
1	1	1	Invalid

CHARACTERISTIC TABLE

Q _n	S	R	Q _{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

EXCITATION TABLE

Q _n	Q _{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0



- **Reflective Critique:**

- ❖ ***Feedback of practice from Students.***

- ✓ Students felt that they understood the concepts very easily and able to reproduce it in a paper
 - ✓ Students felt that activity was very useful and interesting and help to remember the concepts easily

- ❖ ***Benefit of the practice:***

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
 - Learning was made very interesting and the students actively participated in this event.

- ❖ ***Challenges:***

- Few Students were unable to reproduce the concepts immediately
 - Late Bloomers were not able to understand some topics during discussion hours.
 - Duration of this activity is not sufficient, since many students felt that they couldn't complete within the stipulated time

References:

- ❖ <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-one-minute-paper>

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Innovative Practice Description

- **Unit / Topic:** Unit V / VHDL
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 20
- **Activity Chosen:** Flipped Classroom
- **Justification:**

Flipped classroom is a “pedagogical approach in which direct instruction moves from the group learning space to the individual learning space, and the resulting group space is transformed into a dynamic, interactive learning environment where the educator guides students as they apply concepts and engage creatively in the subject matter”

Time Allotted for the Activity: 30 Minutes

- **Details of the Implementation:**

Number of Students – 3

Photographer: one student – Mr. R. Geothisman (interested in photography)

Report Prepared by me.

- Materials were shared with students on the topic through canvas LMS
- Students were given 1 week of time for their preparation
- Each student was given 10 minutes for their presentation, totally 3 students volunteer themselves for this event

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO10	PSO2
C202.1	3	2	1	1

- **PO / PSO mapped:**

Innovative practice	PO1 3	PO2 2	PO10 1	PSO2 1
Justification for correlation	Students apply fundamental mathematical and technical concepts to understand the VHDL Code	Students apply necessary syntax and code procedures to write VHDL Code for the given component	Students discuss with the teacher and effectively communicate their points in front of the class	Students will be able to design and test the electronic system in the engineering applications.

- **Images / Screenshot of the practice:**

A view of implementation of the Activity while the students are sharing the points





- **Reflective Critique:**

- ❖ ***Feedback of practice from Students.***

- ✓ Students felt that they understood the concepts very easily when they discuss and learn with their friends.
- ✓ Students felt that activity was very useful and interesting and help to remember the concepts easily

- ❖ ***Benefit of the practice:***

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- Learning was made very interesting and the students actively participated in this event.

- ❖ ***Challenges:***

- Student has some stage fear while delivering the content and they managed to deliver the content.
- Late Bloomers were not able to understand some topics during discussion hours.

References:

- ❖ <https://www.panopto.com/blog/7-unique-flipped-classroom-models-right/>

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